Teacher : Sakthi, you can do all the shapes you like. But you see, there is enough difficulty in describing shapes made only of lines. So, we will get to curves and circles later.

Sakthi : Teacher, Why should we describe shapes? We can just draw and show them when we want.

Is Sakthi right?



In Mathematics, every thinking is interesting and unlimited. Remember numbers; it is not stopped with adding 2 – digits. Numbers go on endlessly and it is possible to add any two numbers, however large. We know that even a 37- digit number ending in 0 is divisible by 5.

It is the same with shapes too. We are interested in lines, triangles, rectangles and in whatever the shapes may be and the size however large or small. We need to give them names not only to describe them but also to explore a lot with them.

## 4.2 Describing lines

A line can be long or short. A line can be flat or slant or vertical. If a line is rotated in any direction, it remains to be a line. So, given below (Fig. 4.3) are lines in different positions.

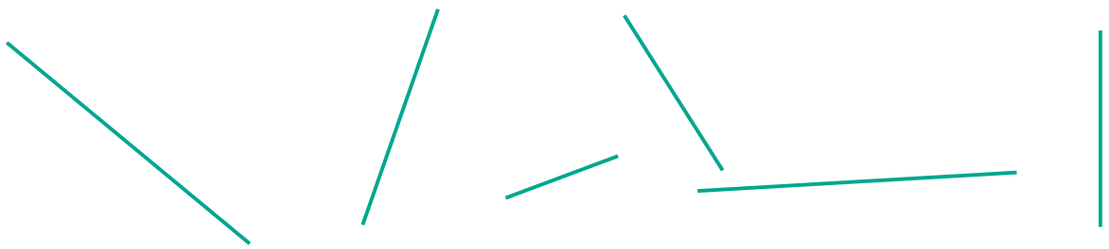


Fig. 4.3

But the following (Fig.4.4) are not straight lines.

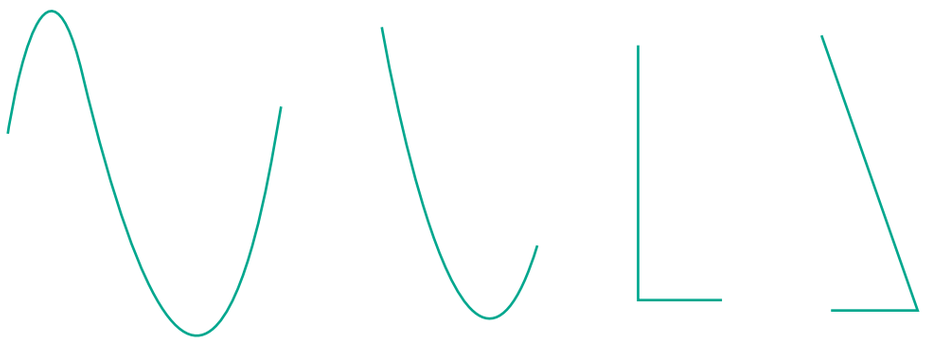


Fig. 4.4

If the length of a line is ignored, then it can be extended in both the directions without ending as in (Fig. 4.5) given below. A line through two points A and B is written as $\overleftrightarrow{AB}$ or $\overleftrightarrow{BA}$. Also it is denoted by a letter ' $l$ '.




Fig. 4.5



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